# Experiment-fed first-event prediction and finite-menu decisions without a fine operating-rate table

Independent constructive route for the acquired joint-event endpoint

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**Abstract**

We construct the missing physical interfaces of a rare first-event generator from faithful local copies, additive fields and priced state pins. Fresh local experiments implement the true Gibbs and operating-chain updates to any paid finite accuracy. Relative probabilities of rare coordinate-pinning events are obtained by heavy completions and locally measured weight ratios; a fine table of energy factors is optional and a fine table of operating rates is unnecessary. The completed static disconnection construction and first-event sampler can consequently be used for the true physical law, with a total variation certificate for the joint quantized event time, terminal state and precursor record. All physical errors are conditional-on-history kernel errors, so reusing an uncontrolled local cell does not assume independent records. Finite-menu bounded-loss decisions follow by martingale concentration. The result consumes fresh experiments on every generated record. It is not an unlimited offline simulator learned from a finite data set. Its premises include faithful local copies and complete local model closure, certified mixing and fixed static contrast, calibrated and reproducible fields, finite detector and timer precision, and paid local state pins. The sufficient cost is polynomial without an inverse global-rarity factor, but may be large.

## 1 Endpoint and accepted machinery

The endpoint is a quantitative law and decision about a first event in a physical system, earned from local experiments without observing its global rare event. The output record is the one defined in the completed first-event construction: a finite quantized event time, the terminal configuration, and a requested finite preceding window of the path. Its error is joint total variation, not relative error for arbitrarily tiny subevents or deadlines. The acquired relative whole-CDF certificate remains a distinct operation.

We use, at their exact accepted scopes:

1. The correlated all-terminal disconnection Count/SampleB construction: positive Gibbs weights, coordinate-pinned sampling, and fixed conditional failure-surprisal contrast. Its operations are cut witnesses, relative masses of coordinate pinnings, pinned draws, and inverse-multiplicity correction.

2. The finite first-event sampler: forward and stationary reversed updates, relative stationary counting, unconditional-TV failure-conditioned sampling, a stationary mass floor and an additive-symmetrization mixing bound. Compilation and a record draw have deterministic finite polynomial caps in the supplied interface costs, inverse gap, log inverse mass floor, requested window length, clock digits, and inverse error. There is no inverse stationary failure probability or mean-lifetime simulation factor.

3. The faithful-cell physical access and paid equilibrium-safe preparation of the accepted active-acquisition construction.

The exact governing sources are the public first-event package at revision

`719e201332992002528c221c49f80f69431485f6`,

especially its first-event report and block completion, active-acquisition certificate, and preserved correlated-reliability report [1]. We construct their interface here; in particular an unknown factor table is not silently treated as a supplied numerical weight oracle.

## 2 Actual access and finite parameters

There are $N$ binary coordinates, known interaction neighborhoods $\partial i$ of maximum degree $d$, and a finite allowed intervention menu $\mathcal{U}$. Write

$$W^u(x) = \exp\left(\sum_i (h_i + u_i)x_i + \sum_{\{i,j\}} J_{ij}x_ix_j\right), \qquad \pi^u(x) = W^u(x)/Z^u, \qquad \ell_i(s) = h_i + \sum_{j\in\partial i} J_{ij}s_j.$$

The fields and couplings are unknown. Known conservative bounds are $|\ell_i(s)| \le B_0$ and $|u_i| \le U$. The operating dynamics are

$$Q^u(x, x^{(i)}) = r_i^u(s)\sigma\big((1-2x_i)(\ell_i(s)+u_i)\big), \qquad 0 < r_* \le r_i^u(s) \le r^*. \tag{1}$$

The clock $r_i^u(s)$ may depend on the neighborhood and on the whole menu environment $u$. It is the same for the two focal-bit values at fixed $(i, s, u)$, as the declared heat-bath form requires. It need not equal a clock observed at an artificial field. These are complete declared mechanisms; finite observations do not infer the absence of external catastrophes. The uniform bounds $r_*, r^*$ must be supplied by a justified operating backbone/specification or actually acquired uniformly over their declared scope. A finite set of sampled contexts does not certify them at unvisited contexts of an arbitrary clock function. The polynomial-degree alternative below relies on these supplied uniform kinetic bounds; it does not infer them from its $D_E$ energy measurements.

A faithful local cell can realize each requested neighborhood $s$, the full operating environment $u$ when used, and the focal two-state process. A held artificial field $z$ has stationary bit probability $q_z = \sigma(\ell_i(s)+z)$ and total relaxation rate at least $r_{\mathrm{cal}} > 0$. The artificial-field clock can otherwise be unknown and field dependent. The local cell may be reused from an arbitrary prior focal state. After a hold $t$, its bit law, conditional on the entire previous transcript, differs from Bernoulli$(q_z)$ by at most $e^{-r_{\mathrm{cal}}t}$. A priced focal pin sets either starting bit before an operating first-exit experiment. The first exit is observed without an unpriced missed-event channel. Response delays, dead times, bit-read errors and timestamp errors must either satisfy the budgets below or be charged as additional conditional error. An absorbing physical event requires a faithful ergodic continuation with the same pre-hit law, as in the accepted construction.

We charge the following actual services as functions, rather than assigning them zero cost:

$$C_{\mathrm{context}},\ C_{\mathrm{field}}(\xi),\ C_{\mathrm{slew}},\ C_{\mathrm{settle}},\ C_{\mathrm{pin}},\ C_{\mathrm{read}}(\beta),\ C_{\mathrm{event}}(\beta),\ C_{\mathrm{timer}}(\Delta).$$

They include context realization, additive-field calibration to absolute accuracy $\xi$, the available slew schedule, settling, state pins, readout/detection and reference timing. Parallel cells change the scheduling, not the sum of exposures or number of pins. A maximum allowed slew rate is not a guaranteed available schedule. The bounds below specify the precision and number of services required; a platform has a finite operation only when it supplies those services at the charged precision.

## 2.1 Coarse structural certification and fine field metrology

The accepted compensating-field search, using bounded conditional-bias observations and martingale Hoeffding, locates a balance field $z_{i,s}$ with true $q_{z_{i,s}} \in [1/4, 3/4]$. It can also supply coarse intervals for $\ell_i(s)$. The number of contexts is $C_0 = \sum_i 2^{|\partial i|}$ unless a proved pairwise reconstruction reduces it. We do not call this polynomial in $d$; it is polynomial at fixed degree.

A stronger pairwise alternative avoids this enumeration when its explicit sufficient certificate succeeds. The accepted learner uses $D_E = N + 2|E_{\text{int}}|$ energy contexts to estimate $h_i$ to error $\eta_0$ and $J_{ij}$ to error $2\eta_0$. Choose $\eta_0 \le [8(1+2d)]^{-1}$ and a paid actual field-setting error at most $1/8$. For every adaptively requested context $s$, $|\ell_i(s) - \widehat{\ell}_i(s)| \le (1+2d)\eta_0 \le 1/8$. Thus the implemented balancing request $z = -\widehat{\ell}_i(s)$ gives $|\ell_i(s) + z| \le 1/4$, placing its actual coin in $[1/4, 3/4]$. The finite energy acquisition is polynomial in $N, d$ and inverse coarse accuracy, rather than a scan of $2^d$ contexts. The sufficient all-field, all-pinning influence certificate

$$\max_i \sum_{j \in \partial i} \tanh\big((|\widehat{J}_{ij}| + 2\eta_0)/4\big) \le 1 - \alpha$$

is valid because the maximum variation of a logistic conditional when its logit changes by $J$ is $\tanh(|J|/4)$. Interval sums of positive and negative coupling endpoints give full conditional bounds for each menu environment. This certificate can be stricter than exact context enumeration; both options are retained. Neither option acquires a table of $r_i^u(s)$: adaptively encountered operating contexts are handled directly by the censored-exit experiment.

The intervals must certify a pinned Dobrushin contraction margin $\alpha > 0$ for every menu environment and full conditional failure-probability bounds

$$0 < q_i^- \le \pi^u(x_i = 1 \mid x_{-i}) \le q_i^+ < 1, \qquad q_i^- \ge (q_i^+)^c \tag{2}$$

with fixed $c$. Coarse interval propagation provides such a certificate if the true system has strict margins and refinement reaches them. With full conditional logit error at most $\nu$, each probability changes by at most $\nu/4$, hence each row's influence uncertainty is at most $d\nu/2$. For $a \in \mathbb{R}$, the derivative of $\log[-\log \sigma(a)]$ has absolute value at most one; thus a true surprisal contrast $c_0$ is certified by $c$ if $c_0 e^{2\nu} \le c$, using the corresponding worst-case bounds. All refinements have declared caps. Failure to certify by a cap returns *premises unresolved* and is not evidence that the premise holds. Supplied valid certificates may instead be used.

This coarse localization does *not* eliminate fine physical metrology. The Gibbs coin below uses the actual shift $u - z$. To achieve absolute coin error $\theta$, the additive fields used in that transformation must be known and reproduced to $O(\theta)$, costing $C_{\text{field}}(O(\theta))$. Likewise deployment must reproduce the chosen actual field environment. Without arbitrary required precision, the construction stops at the corresponding attainable error floor.

For convenient conservative common bounds define

$$b_E = \lceil 2(B_0 + U) \rceil, \qquad \zeta = 2^{-N(b_E+1)}, \qquad \rho = 2r^*, \quad R = N\rho, \quad g = \alpha r_*/R. \tag{3}$$

Every full conditional bit probability is at least $\sigma(-(B_0+U)) \ge 2^{-(b_E+1)}$, and successive conditioning preserves this bound by the tower property. Thus $\pi^u(x) \ge \zeta$. The operating generator dominates the heat-bath Dirichlet form with rate $r_*$, whose certified gap is at least $\alpha$ in the accepted normalization. Consequently $P = I + Q^u/R$ has gap at least $g$. Notice that $\rho$ uses *twice the upper clock bound* $r^*$, not $r_*$.

## 3 A physical true-Gibbs update without fine energy estimation

**Lemma 3.1** (Balanced two-coin construction)**.** *Suppose a balanced physical coin has success probability $q = \sigma(\ell+z) \in [1/4, 3/4]$. For $a = e^{u-z}$ the following experiment has output law* Bernoulli($\sigma(\ell+u)$)*: choose branch 1 with probability $a/(1+a)$ and branch 0 otherwise; toss a $q$ coin; on branch 1 accept a success and return 1; on branch 0 accept a failure and return 0; otherwise repeat. The acceptance probability on each attempt is at least $1/4$.*

*Proof.* The probabilities of accepted 1 and accepted 0 are $aq/(1+a)$ and $(1-q)/(1+a)$. Their sum is at least $\min(q, 1-q) \geq 1/4$ and their normalized ratio is $aq/(aq+1-q) = \sigma(\ell+u)$. □

This is a two-coin/Bernoulli-race identity, established machinery [3, 4]; the new use here is its physical interface and composition with rare-event generation.

For a requested $0 < \theta \leq 1/4$, set

$$J = 4\lceil \log_2(8/\theta) \rceil, \qquad \beta = \theta/(16J).$$

Cap the experiment after $J$ attempts, returning a fixed fallback bit on exhaustion. The ideal tail is at most $(3/4)^J \leq e^{-J/4} < \theta/8$. Before each read hold for $t_{\mathrm{hold}} \geq r_{\mathrm{cal}}^{-1} \log(1/\beta)$. This makes the physical coin within conditional TV $\beta$ of $q$. A finite dyadic branch selector has error at most $\beta$. The two substitutions over at most $J$ attempts add at most $2J\beta = \theta/8$.

Let the numerical values of each of the actual fields $u, z$ be within $\xi \leq \theta/4$. The computed log shift then differs by at most $2\xi$; since $\sigma$ is 1/4-Lipschitz, the resulting infinite-factory output probability differs by at most $\xi/2 \leq \theta/8$. Acceptance remains at least $1/4$ for every positive branch weight even with the perturbed shift. Finite exponential/selector evaluation is included in the selector error. Thus the entire capped output has conditional TV at most $\theta$, with slack for the declared response/readout error allocations. Standard interval exponentials and dyadic thresholds require bit cost polynomial in $B_0 + U + \log(J/\theta)$; their cost is not proportional to $e^{B_0+U}$.

The actual service bill per requested Gibbs update is bounded by

$$\begin{aligned} C_G(\theta) \leq\ & C_{\mathrm{context}} + C_{\mathrm{field}}(\theta/4) + C_{\mathrm{slew}} + C_{\mathrm{settle}} \\ & + J\big[r_{\mathrm{cal}}^{-1} \log(16J/\theta) + C_{\mathrm{read}}(\beta) + C_{\mathrm{timer}}(\Delta_G)\big] + C_{\mathrm{arith}}. \end{aligned} \tag{4}$$

The hold-timing resolution $\Delta_G$ is chosen so the actual hold is at least the specified time; rounding upwards suffices if its excess time is charged. This method requires no focal-state reset because its bias bound holds uniformly conditionally on every earlier experimental history.

**Lemma 3.2** (Balanced log-odds measurement)**.** *At a balanced context, a finite number of physical readings, $O(h^{-2}\log(1/\beta_0))$, and additive-field calibration to $O(h)$ estimate $\ell_i(s)$ to absolute error $h$, with conditional failure probability at most $\beta_0$. Every reading holds for $O(r_{\mathrm{cal}}^{-1}\log(1/h))$. All costs are uniform over the previous experimental transcript.*

*Proof.* For $h \leq 1$, set $a = h/128$. Hold until each conditional mean differs from the true $q_z$ by at most $a/2$. Martingale Hoeffding on $M \geq 2a^{-2}\log(2/\beta_0)$ observations bounds the empirical error by $a$, including the conditional-mean bias, with probability at least $1-\beta_0$. Clip the empirical probability to $[1/8, 7/8]$. Since true $q_z \in [1/4, 3/4]$, clipping cannot increase its error. The logit derivative on the clipping interval is at most $64/7 < 10$, so its error is at most $10a < h/12$. Subtract the measured field $z$, whose actual value is known to $h/4$, and evaluate the logarithm to $h/4$. The total is less than $h$. For $h > 1$ one may use the $h = 1$ procedure. The hold and every calibration, response, read and timer service are charged. □

No fine log-odds estimate is used in the Gibbs update of Lemma 3.1. Lemma 3.2 is used only when an actual relative static probability needs a local weight-ratio evaluation.

# 4 True operating updates from censored exits

The actual operating clocks can change with the menu. We never infer them from the clocks at balancing fields. Fix a requested coordinate, its neighborhood, starting bit, and full operating environment $u$. Pin that bit, release, and let its first-exit time be $T \sim \mathrm{Exp}(\lambda)$, where $\lambda$ is the applicable rate in (1). It obeys $0 < \lambda \le r^*$.

**Lemma 4.1** (Operating race and finite walk)**.** *Take $\rho = 2r^*$, generate an independent $E \sim \mathrm{Exp}(\rho)$, and censor the focal first-exit observation at $E$. The bit $\mathbf{1}_{\{T<E\}}$ has probability $p = \lambda/(\lambda+\rho) \le 1/3$. A random walk starting at 0, moving up 1 on a $p$-coin success and down 1 otherwise, ever hits $+1$ with probability $p/(1-p) = \lambda/\rho$. For $0 < \theta \le 1/4$, put*

$$m = \lceil \log_2(8/\theta) \rceil, \qquad n = 72m.$$

*Stop the walk at $+1$, at $-m$, or after $n$ steps; output 1 only if $+1$ has been reached. Its output probability differs from $\lambda/\rho$ by less than $\theta/4$. If each implemented input race bit has conditional TV error at most $\beta = \theta/(8n)$, the full output error is less than $\theta/2$.*

*Proof.* Independence of the two clocks gives $\mathbb{P}(T < E) = \int_0^\infty \lambda e^{-(\lambda+\rho)t}\, dt$. The infinite-walk hitting probability is the classical biased-walk solution $r = p/(1-p)$, obtained by solving the bounded harmonic hitting equation or optional stopping on finite intervals and taking a limit. Since $r \le 1/2$, the paths lost by absorption at $-m$ have probability at most the eventual success probability starting there, namely $r^{m+1} \le 2^{-m-1} < \theta/16$.

For the unlimited walk, $\mathbb{E}S_n \le -n/3$. A path not yet absorbed at either boundary has $S_n > -m$. Since $n \ge 6m$, Hoeffding for independent increments in $[-1, 1]$ gives

$$\begin{aligned} \mathbb{P}(\text{not absorbed at } n) &\le \mathbb{P}(S_n - \mathbb{E}S_n > n/3 - m) \\ &\le \exp[-(n/3-m)^2/(2n)] \\ &\le e^{-n/72} = e^{-m} \le 2^{-m} \le \theta/8. \end{aligned}$$

The reference walk uses independent ideal coins. Conditional kernel errors of the physical implementation are coupled sequentially: the probability any of at most $n$ races differs is at most $n\beta = \theta/8$. Adding the three errors proves the result without assuming the physical coins independent. □

The construction is a particular Bernoulli factory; the general analytic Bernoulli-factory theory is established [2]. The explicit bounded walk, censored physical race and total error/cost accounting are provided because they are needed by the claimed physical operation.

## 4.1 A finite deadline and comparison device

For each ideal race input at accuracy $\beta$, a sufficient finite recipe is

$$D = \rho^{-1} \lceil \log_2(32/\beta) \rceil, \qquad 0 < \Delta \le \beta/(256\rho). \tag{5}$$

The ideal deadline tail is $e^{-\rho D} \le \beta/32$. Use an independent uniform $b$-bit grid variable to implement the inverse exponential map. Couple it to a true $U \in (0, 1)$ by a uniform grid interval. On $E = -\log U/\rho \le D$ the map's derivative is at most $e^{\rho D}/\rho$. A sufficient uniform grid mesh is

$$2^{-b} \le \min\{\beta/256,\ \rho\Delta e^{-\rho D}/32\}.$$

An endpoint cell intersecting $e^{-\rho D}$ has probability at most $2^{-b}$ and is charged separately. The endpoint $U = 0$ is never evaluated: all grid cells below the cap threshold return the declared fallback. On the other cells, finite interval evaluation and rounding produce a scheduled deadline within $\Delta$ of the coupled ideal $E$. Choosing interval width at most $\Delta/4$ and rounding to a clock grid of spacing at most $\Delta/4$ supplies this bound. The needed bit length is $O(\log(1/\beta) + \log(1/(\rho\Delta)))$; the displayed inequalities can be checked with rational conservative exponential bounds.

Include actual deadline service error and timestamp/response errors in a combined comparison error of at most $3\Delta$. The physical comparison can then differ from the ideal event only if $|E - T| \leq 3\Delta$, or on a cap or numerical failure. Conditional on $T$, the density of the independent $E$ is at most $\rho$, so the tie band has probability at most $6\rho\Delta$. Use a fixed rule in ties. Thus cap, threshold cell, rounding and tie-band errors sum to less than $\beta/8$ with the choices above; the remaining budget up to $\beta$ covers declared read, missed-exit and service errors. If these errors are deterministic accuracy bounds rather than random failures, their contributions are just the tie-band bound. No total variation approximation of a continuous real clock by a discrete variable is asserted; only the event-comparison bit is approximated.

Each race ends by $D + O(\Delta)$, including a no-exit observation. It never waits for an uncontrolled rare full dwell $1/\lambda$. But it uses a new requested-state pin and release before every trial, even when that state is rarely occupied naturally. The pin is a substantive resource.

### 4.2 Full operating kernel and cost

Choose a coordinate uniformly from $N$, request its actual operating environment and neighborhood, and use Lemma 4.1 to flip it with probability $\lambda_i/\rho$. The ideal kernel is exactly $P = I + Q^u/(N\rho)$. A finite coordinate selector can use an error budget $\theta/4$; the preceding factory has slack for the remaining budget.

For the actual source and target states, $\pi^u(x)Q^u(x, x^{(i)}) = \pi^u(x^{(i)})Q^u(x^{(i)}, x)$, because the same $r_i^u(s)$ multiplies the complementary logistic probabilities. Therefore $P^* = P$. This route acquires both update interfaces only for the declared reversible form. It does not construct the reversed kernel of an arbitrary unknown nonreversible process.

A worst-case service cost for an operating update at error $\theta$ is

$$\begin{aligned} C_P(\theta) \leq\ & C_{\text{context}} + C_{\text{field}}(\xi_{\text{op}}) + C_{\text{slew}} + C_{\text{arith}} \\ & + n\big[C_{\text{pin}} + C_{\text{settle}} + C_{\text{event}}(\beta) + C_{\text{timer}}(\Delta) + D + O(\Delta)\big], \end{aligned} \tag{6}$$

with $m, n, \beta, D, \Delta$ as above and with any operating-environment reproduction precision $\xi_{\text{op}}$ required to produce the declared actual $Q^u$. When faithful copies reproduce that environment exactly by assumption, $C_{\text{field}}(\xi_{\text{op}})$ denotes that paid reproducible setting. Otherwise the induced kernel difference is separately bounded and allocated to $\theta$; an unknown arbitrary dependence of $r_i^u$ on the field does not permit a continuity inference from field accuracy. Deployment must reproduce the same environment or carry an additional proved law-error bound.

The number of pins is $O(\log(1/\theta))$, and the total censored dwell is $O(\rho^{-1}\log^2(1/\theta))$ with the explicit conservative constants above. Small rates need not be estimated: a small success probability is implemented by a correspondingly unlikely success. This does not improve the cost of *estimating* a small rate relatively. It does not remove the numerical $r_*^{-1}$ dependence of $g^{-1}$ or the cost of physically setting a locally unlikely starting bit.

# 5 Experimental rare static access without an unknown weight oracle

**Lemma 5.1** (Finite experimental pinned sampler)**.** *For any coordinate pinning $D$ and $0 < \omega \le 1/4$, a conditional-on-history TV-$\omega$ draw from $\pi^u(\cdot \mid D)$ uses at most $NS$ Gibbs updates, where*

$$S = \left\lceil \alpha^{-1} \left\lceil \log_2(4N/\omega) \right\rceil \right\rceil, \qquad C_{\mathrm{pinlaw}}(\omega) \le NS\, C_G\big(\omega/(4NS)\big) + C_{\mathrm{virtual}}.$$

*The pin constraints here are virtual classical constraints. They are not a claim that the whole physical system can be prepared in the rare pin event.*

*Proof.* Start a virtual configuration in any pin-compatible state. Systematically update its free coordinates, requesting a local cell at the current virtual neighborhood each time. Pinning preserves the influence bounds. For ideal Gibbs updates, if all coordinate disagreement probabilities before a sweep are at most $d_0$, none exceeds $d_0$ during it and each updated coordinate has disagreement at most $(1-\alpha)d_0$. A complete sweep gives that bound everywhere. Coupling to a stationary copy gives TV at most $N(1-\alpha)^S \le Ne^{-\alpha S} \le \omega/4$. Substituting conditional-TV-$\omega/(4NS)$ physical updates costs at most $\omega/4$ by sequential coupling. The result has slack for finite control and any charged services. If all coordinates are pinned, return that state. □

The accepted heavy-prefix estimator used the numerical terminal weight $W(x_D)$. It cannot be copied unchanged when $W$ is unknown. The following identity closes precisely that gap.

**Lemma 5.2** (Experimental relative pin mass)**.** *For any nonempty coordinate-pinning event $D$, rational $0 < v, \beta_0 \le 1/4$, the physical access constructs a positive rational $\widehat{p}_D$ with*

$$\mathbb{P}\big(e^{-v}\pi^u(D) \le \widehat{p}_D \le e^{v}\pi^u(D) \,\big|\, \textit{prior transcript}\big) \ge 1 - \beta_0.$$

*Its worst-case number of pinned draws is $O(N^3 v^{-2}\log(N/\beta_0))$; each draw has TV tolerance $O(v/N)$. It additionally uses $O(N^3 v^{-2}\log(N/\beta_0))$ balanced readings to estimate local log odds, at field precision $O(v/N)$ and hold time $O(r_{\mathrm{cal}}^{-1}\log(N/v))$. There is no inverse $\pi^u(D)$ cost.*

*Proof.* Heavy-complete $D$ by visiting its free coordinates. At each stage estimate the current conditional marginal and choose the empirically heavier bit, then add that bit to the prefix. Independently in the algorithmic sense (fresh future experiments, not a physical reset assumption), heavy-complete the empty pin event. Denote their terminal states by $x_D, x_0$, and the products of their true selected conditional probabilities by $H_D, H_0$. The exact identities

$$H_D = \frac{\pi^u(x_D)}{\pi^u(D)}, \qquad H_0 = \pi^u(x_0)$$

give

$$\pi^u(D) = \frac{W^u(x_D)}{W^u(x_0)}\,\frac{H_0}{H_D}. \tag{7}$$

When $D$ pins every coordinate, $H_D = 1$ and the same identity applies.

Put $a = v/(32N)$. Estimate each of at most $2N$ marginal stages using

$$M \ge 2a^{-2}\log(8N/\beta_0)$$

pinned draws, each at conditional TV at most $a/2$. Conditional mean bias is at most $a/2$, and bounded martingale Hoeffding bounds the empirical deviation around its conditional means by $a/2$

with failure probability at most $\beta_0/(4N)$. A union bound gives a simultaneous event of probability at least $1-\beta_0/2$. On it, every empirically selected true heavy probability obeys $h_j \geq 1/2-a \geq 3/8$ and its estimated value is at least $1/2$. The log error of one factor is at most $(8/3)a$ by the mean-value theorem. The two products thus contribute at most $(16/3)Na = v/6$ to the log error in (7).

Connect $x_0$ to $x_D$ by changing their differing coordinates one at a time, in any fixed order; there are at most $N$ steps. If step $j$ changes coordinate $i_j$ of $x^{j-1}$, the log weight ratio is exactly

$$\log \frac{W^u(x_D)}{W^u(x_0)} = \sum_j (1-2x_{i_j}^{j-1})\big[\ell_{i_j}(x_{\partial i_j}^{j-1}) + u_{i_j}\big]. \tag{8}$$

These are full local conditional log odds, not marginals under the rare pin event. Estimate each bracket to accuracy $v/(4N)$ with Lemma 3.2, including the actual $u$ calibration and numerical evaluation in that tolerance. Allocate failure $\beta_0/(2N)$ per term. Their total log error is at most $v/4$ on a joint event of failure at most $\beta_0/2$. The chosen contexts are adaptive, but the conditional guarantees are uniform, so the union bound remains valid.

Compute a finite positive rational exponential of the resulting sum to log-relative error at most $v/12$, and multiply by the rational empirical heavy products. The total log error is at most $v/6 + v/4 + v/12 = v/2 < v$, leaving slack. All selected empirical factors are at least $1/2$. The absolute magnitude of (8) is at most $N(B_0+U)$, so the rational output and computations have bit length polynomial in $N(B_0+U)$, $\log(1/v)$, and the sample-count logarithms. There are at most $2N$ heavy stages, each using $M = O(N^2 v^{-2}\log(N/\beta_0))$ pinned draws. At most $N$ balanced logit estimates have the same order of individual readings. This proves the claimed counts. □

One explicit cost form, with the constants of the preceding proof, is

$$\begin{aligned} C_{\mathrm{mass}}(v,\beta_0) \;&\leq 2NM\, C_{\mathrm{pinlaw}}(v/(64N)) \\ &\quad + N\, C_{\mathrm{odds}}(v/(4N), \beta_0/(2N)) + C_{\mathrm{ratio-arith}}, \quad M = \left\lceil 2(32N/v)^2 \log(8N/\beta_0)\right\rceil . \end{aligned} \tag{9}$$

Here $C_{\mathrm{odds}}$ includes every actual calibration, setting, slew, settling, hold, read and timer call in Lemma 3.2.

### 5.1 From pin masses to disconnection Count and SampleB

Let the physical failure event $B$ be all-terminal disconnection of the declared graph, with bit 1 denoting an edge failure. The full failure rule is supplied. The underlying graph is connected and has at least two vertices, so $B$ is nonempty and the cut weights have a positive minimum. The coarse rational bounds (2), with fixed $c$, yield rational cut weights as in the accepted correlated construction. For $C = \lceil 2c \rceil$, that construction has two branches. If its minimum cut weight $\lambda_{\mathrm{cut}}$ is below $4\lceil \log_2(2v_{\mathrm{graph}})\rceil$, it supplies the explicit lower bound $p_0 = 2^{-8C\lceil \log_2(2v_{\mathrm{graph}})\rceil}$ on $\pi^u(B)$; ordinary Gibbs rejection then costs at most $v_{\mathrm{graph}}^{O(C)}$. In the other branch, at target error $h$, it constructs at most

$$J_B \leq 64 v_{\mathrm{graph}}^{4C+6}/h$$

cut events and omits at most $(h/16)\pi^u(B)$ in mass on its declared enumeration event.

Use Lemma 5.2 to estimate each cut-pinning probability $p_j$ by a positive rational $\phi_j$ with $e^{-s}p_j \leq \phi_j \leq e^s p_j$. Draw index $j$ proportional to $\phi_j$, then draw from the pinned law with Lemma 5.1. For exact component draws, the proposal density is

$$\nu(x) = \frac{\pi^u(x)}{\sum_j \phi_j} \sum_j \frac{\phi_j}{p_j} \mathbf{1}_{E_j}(x).$$

With $d(x) = \sum_j \mathbf{1}_{E_j}(x)$, the ratio of the sum in this expression to $d(x)$ lies in $[e^{-s}, e^s]$. The reciprocal statistic $1/d(x) \in [1/J_B, 1]$ therefore gives relative union counting by a bounded-mean estimator; accepting with probability $1/d(x)$ gives the union-conditioned law with a multiplicative density error $e^{\pm 2s}$. This is the exact place where relative pin mass, rather than knowledge of $W$, is needed.

For counting at relative error $h$, the accepted schedule uses $s = h/16$, joint pin-mass confidence error at most one quarter of the requested count failure probability, and at most

$$\lceil 2048 J_B h^{-2} \lceil \log_2(8/\beta_0) \rceil \rceil$$

union proposals. For a conditional draw at TV error $h$, it uses a fresh list, total enumeration/mass failure at most $h/8$, $s = h/32$, and at most $\lceil 2J_B \lceil \log_2(16/h) \rceil \rceil$ proposals. It always returns a state in $B$, using the known all-failed state as the finite-cap fallback. Finite categorical and reciprocal arithmetic are the accepted finite rational procedures. All requested relative pin accuracies and confidences are thus explicit functions of $h, \beta_0, J_B$.

Dependent physical calls are handled by conditional means, rather than requiring every sample to succeed on a shared event. Condition on the good retained-list and mass-estimate event. For the count statistic $Y = 1/d(X)$, every subsequent conditional mean differs from the ideal mixture mean by at most the proposal-TV and rounding budget $h/(64J_B)$. The ideal mean is at least $1/J_B$. Conditional bounded-variable exponential-moment concentration, proved in Lemma 6.2, therefore gives the same relative concentration schedule, with confidence entering only logarithmically. For SampleB, every proposal conditional on every previous history is within its allocated TV of the same ideal proposal. Its accepted mark law is uniformly close to the same union-conditioned law by Lemma 6.3. A mixture over the possible first accepted histories retains that bound, and the cap has the same exponential tail. Thus the accepted static tolerance schedule is valid with fresh conditional-on-history physical calls, without an iid assumption. Running it at half the requested error supplies slack for conservative constants. Pin-mass confidence failures for SampleB are included in its unconditional TV budget; no union over every future SampleB invocation is demanded.

Consequently the exact interfaces needed by the accepted first-event sampler are now physical finite operations:

$$\mathrm{Count}(h, \beta_0), \qquad \mathrm{SampleB}(h),$$

with conditional-on-history confidence/TV guarantees for the true law. A fresh SampleB call rebuilds its random estimates. Reusing a shared random table instead requires an explicitly allocated outer confidence event; it is not an independent sample by declaration.

### 5.2 Optional fitted-static alternative

A distinct implementation may use the accepted active learner to construct a rational static weight table $\widehat{W}$ satisfying, on one shared confidence event, $e^{-E} \le \widehat{W}(x)/W(x) \le e^E$ for every $x$. Then

$$e^{-2E} \le \frac{\widehat{\pi}(B)}{\pi(B)} \le e^{2E}, \qquad \mathrm{TV}\big(\widehat{\pi}(\cdot \mid B), \pi(\cdot \mid B)\big) \le \tanh E.$$

Indeed normalization on the whole space or on $B$ changes a pointwise weight ratio by at most another factor $e^E$, and a likelihood ratio in $[e^{-2E}, e^{2E}]$ has TV at most $\tanh E$. Choose $E$ to spend at most half the requested count/sample error; the known-model static construction supplies the other half. Its pinned mixing and fixed contrast must be certified for the fitted weights too. True physical $P = P^*$ updates still use the operating-exit factory, so no fine kinetic table is needed. This

hybrid can amortize an acquired static table over many records. The on-demand weight-ratio route is an alternative, not a universal cost improvement over that hybrid.

# 6 Adaptive composition and the full acquired operation

**Lemma 6.1** (Predictable physical-query substitution)**.** *Consider a finite randomized program issuing at most $D$ adaptive queries. Suppose that, conditional on every possible preceding experimental transcript, its actual response kernel at query $j$ differs from the specified ideal response kernel by at most $\theta_j$ in TV. If the predictable budgets satisfy $\sum_{j\le D}\theta_j \le a$ deterministically, the complete actual transcript differs from the ideal transcript by at most $a$ in TV. A bad-calibration event of ideal probability at most $\delta_0$ then has actual probability at most $\delta_0 + a$.*

*Proof.* Couple the program's classical randomness identically. Until its first unequal response, its requested inputs agree; couple that pair of kernels maximally. The conditional probability of a first mismatch at query $j$ is at most $\theta_j$. A union bound gives the result. This is a statement about conditional kernels, so no independence of actual cells or their past states is required. □

Every nested routine above has deterministic caps. The conditional concentration estimates and sample/call error budgets can therefore be composed adaptively or flattened into a finite controller. The next two facts retain logarithmic confidence dependence; coupling every high-level calibration sample at error $\delta/D$ can needlessly force inverse-$\delta$ static accuracies.

**Lemma 6.2** (Conditional bounded-mean concentration)**.** *Let $Y_j \in [0,1]$ and let their predictable means satisfy $|\mathbb{E}(Y_j \mid \mathcal{F}_{j-1}) - \mu| \le b$, where $\mu \ge \mu_0 > 0$. For $0 < \eta \le 1/4$, if $b \le \eta\mu_0/16$ and*

$$n \ge \frac{256}{\mu_0\eta^2}\log(8/\delta_0),$$

*then $\mathbb{P}(|n^{-1}\sum_j Y_j - \mu| > \eta\mu/2) \le \delta_0$. No independence or common actual predictable mean is needed.*

*Proof.* Set $\mu_- = \mu - b$ and $\mu_+ = \mu + b$. Convexity on $[0,1]$ gives, for any real $t$, $\mathbb{E}(e^{tY_j} \mid \mathcal{F}_{j-1}) \le 1 + (e^t - 1)\mathbb{E}(Y_j \mid \mathcal{F}_{j-1})$. For $t > 0$ this is at most $\exp(\mu_+(e^t - 1))$, and for $t < 0$ at most $\exp(\mu_-(e^t - 1))$. Iteration and exponential Markov give the usual upper and lower multiplicative Chernoff bounds with $\mu_+$ and $\mu_-$, respectively. At relative deviation $\eta/4$, their failure probabilities sum to at most $2\exp[-n\mu_0\eta^2/96]$ since $\mu_- \ge \mu_0/2$. The stated $n$ is more than sufficient. The deterministic mean bias $b$ plus those deviations is below $\eta\mu/2$. □

**Lemma 6.3** (Conditional accepted records)**.** *Let an ideal single attempt output either rejection or an accepted mark, with success probability $q \ge q_0 > 0$ and successful-mark law $\nu$. At every attempt and every prior history suppose the actual combined rejection/mark law is within TV $a \le q_0/2$ of that same ideal law. Its success probability is at least $q_0/2$, and its conditional successful-mark law is within $2a/q_0$ of $\nu$. After a cap of $J$ attempts, returning any mark on exhaustion gives overall mark TV at most $2a/q_0 + e^{-q_0 J/2}$.*

*Proof.* The success probabilities differ by at most $a$. Subtract the normalized accepted subprobabilities and add and subtract the actual subprobability divided by the ideal $q$. The numerator difference costs at most $a/q$ and its normalization costs at most $a/q$. This bounds conditional TV by $2a/q_0$. It is uniform over each pre-success history, hence also over their mixture. Conditional failure probability at any attempt is at most $1 - q_0/2$, so exhaustion is at most $(1 - q_0/2)^J$. Neither assertion assumes independent physical attempts. □

Here is an explicit allocation for one intervention and one compiled record generator, for $0 < \varepsilon, \delta \le 1/4$.

1. Allocate $\delta/4$ to coarse structural certification, balancing, and any shared metrology confidence. The declared model/faithfulness premises themselves are not statistical hypotheses certified by this event.

2. Run the accepted first-event compilation at output target $\varepsilon/2$ and confidence failure $\delta/2$, using the true static interfaces just constructed. Include every pin-mass and cut-enumeration confidence in its requested interface budget. Use its deterministic finite controller and clamp failed estimates to the same certified floors/ranges as the accepted construction.

3. For the Palm calibration mean, use the accepted $\xi = (\varepsilon/2)/128$, $q_0 = g/8$ and per-proposal TV budget $\rho_{\mathrm{prop}} = \xi q_0/64$. Include finite acceptance rounding so every observable's conditional mean bias is at most $2\rho_{\mathrm{prop}} \le \xi q_0/16$. Use $n_{\mathrm{Palm}} \ge 256(q_0\xi^2)^{-1}\log(32/\delta)$. Lemma 6.2 supplies the required relative Palm-mean confidence. Count has the same conditional bounded-mean construction. Each calibration proposal's accuracy depends on $\varepsilon, g$ and its finite local length, not on $\delta/n_{\mathrm{Palm}}$.

4. Let $D_1$ be the deterministic cap for one later record. Allocate total substitution error $\varepsilon/4$, for example $\theta_1 = \varepsilon/[4\max(1, D_1)]$ per query. For a nested module, split its allocated query error over its own finite calls. The requested error is the smaller of its own proof's tolerance and the outer allocation.

5. Supply the requested initial law to TV $\kappa \le \varepsilon/4$, or an exact initial state. All finite-bit clock and record-rounding errors already paid by the accepted sampler remain in its $\varepsilon/2$ budget.

All controller caps are computed before execution from the accepted explicit schedule and the displayed static caps. To avoid a circular precision definition, first expand only down to ideal Gibbs-update, operating-update and balanced-coin requests. Their number is fixed by the static sample counts and mixing sweep lengths. Replacing a request by its finite physical factory changes its service cost, not that outer query count. An outer tighter budget changes hold length or factory cap, not the number of observations already specified by a mean estimator. Module-wise allocation using Lemma 6.3 is a sharper valid alternative to the per-record coupling: it gives the accepted conditional path error $4\rho_{\mathrm{prop}}/q_0$ and cap tail directly.

**Theorem 6.4** (Experiment-fed true first-event generator)**.** *Under the declared model and physical services of Section 2, with fixed $c$, a valid pinned contraction margin $\alpha$, and the all-terminal disconnection failure rule, the preceding finite controller constructs a compiled generator with the following property. With probability at least $1 - \delta$ over its shared calibration, for every past experimental transcript and every efficiently supplied initial law $\mu$, its next record has conditional law within $\varepsilon$ in total variation of the true quantized first-event record from $\mu$ for $Q^u$. The ideal initial law is the target of the supplied sampler; its preparation/draw error is charged as above. The generator returns the event time, terminal configuration, and requested preceding window jointly.*

*Its operation uses fresh faithful local experiments, finite bits and bounded physical service times on every draw. It needs no fine table of operating rates. The on-demand static implementation needs no fine exported energy-factor table, while the fitted-static hybrid is also valid. Its full classical and local-query counts are polynomial in the graph/input description at fixed $c$ and either fixed degree or the successful pairwise coarse certificate above, $B_0 + U$, $\alpha^{-1}$, $g^{-1}$, requested window length, clock digits, inverse output accuracy, and log inverse confidence, including the coarse certification*

*refinement cap. The field, pin, setting, response, timer, memory and propagation costs are the actual charged service functions at the resulting tolerances. There is no inverse global stationary-failure probability or mean-lifetime trajectory factor.*

*Proof.* For ideal physical response kernels, Lemmas 5.1 and 5.2, the exact overlap identities, and their finite confidence/precision schedules supply Count and SampleB for the true invariant law. Lemma 4.1 supplies true forward and reversed $P = P^*$ updates. Equation (3) supplies the mass floor and gap. The hypotheses of the accepted first-event computation theorem therefore hold for this same true $Q^u, \pi^u, B$.

The mathematical path, Palm and mixing arguments of that theorem do not require rates to be rational. Rationality in its numerical formulation allows finite evaluation of update probabilities. The physical finite kernel implementations above replace that evaluation. Known bounds, requested precisions and the controller are rational, and every returned record remains finite. Unknown real parameters are not evaluated by a hidden real-arithmetic oracle.

In compilation, the true-law Count confidence and Lemma 6.2 for each bounded Palm statistic retain total calibration failure at most $\delta/2$ with the displayed conservative sample counts. Their implementation bias is already included. The shared structural event adds at most $\delta/4$, leaving slack. Thus with probability at least $1 - \delta$ the realized compiled parameters have their true-law correctness property. Conditional on any such compiled state and any later experimental transcript, Lemma 6.1 bounds each record's additional physical substitution error by $\varepsilon/4$. The accepted record error is $\varepsilon/2$, and the initial-law error is at most $\varepsilon/4$. These add to $\varepsilon$. Uniformity of the physical conditional kernels makes the per-record bound valid after reusing cells or after any earlier record outputs. It does not assert unconditional independence of the physical records.

The accepted controller's finite caps have no inverse global-rarity term. The constructed static schedule has $v_{\mathrm{graph}}^{O(c)}$ cut complexity and polynomial inverse requested accuracy, while a local operating update uses $O(\log(1/\theta))$ censored exits. The static pin-mass identity never waits to see the rare pin event. Substitution therefore inserts only the displayed polynomial query counts and actual service functions; no hidden $1/\pi^u(B)$ or $\mathbb{E}_\mu \tau$ is introduced. □

The confidence event is shared across all later draws from a compiled generator. Fresh record errors are conditional distribution errors, not additional claims that every physical trial succeeds with certainty. If random metrology failures are allowed on fresh calls, their probabilities must be included in the per-record budgets. An indefinite sequence of draws does not receive a zero-failure guarantee without a summable allocation. This is a true-law *generative physical procedure*, not an unlimited offline algorithm learned once and then run without further experiments.

### 6.1 A quantitative physical ledger

Let $M_{0,G}, M_{0,P}, M_{0,O}$ denote compilation caps for Gibbs updates, operating updates and balanced log-odds estimates after expanding all static routines, and $M_{1,G}, M_{1,P}, M_{1,O}$ the corresponding per-record caps. They are obtained directly from the preceding displayed caps and the accepted first-event schedule. For prescribed allocations $\theta_{a,j}$ and log-odds accuracies/confidences $(h_{a,j}, b_{a,j})$, the exact cost ledger is

$$\begin{aligned} C_a \le C_{a,\mathrm{classical}} &+ \sum_{j=1}^{M_{a,G}} C_G(\theta_{a,j}) + \sum_{j=1}^{M_{a,P}} C_P(\theta_{a,j}) \\ &+ \sum_{j=1}^{M_{a,O}} C_{\mathrm{odds}}(h_{a,j}, b_{a,j}), \qquad a = 0, 1. \end{aligned} \tag{10}$$

Compilation additionally pays coarse acquisition and shared metrology. All storage for configurations, cut lists, counters, finite clock words, and the precursor window is the accepted polynomial classical memory; storing the earlier entire long physical trajectory is unnecessary. Apparatus size, available cells, propagation and scheduling determine the wall-clock realization of the sum. The charged $C_{\mathrm{timer}}$ includes reference-frequency calibration and drift sufficiently accurate over the whole interval $[0, D]$ to meet (5), not just a nominal digital timestamp spacing. A requested physical precursor duration $h_{\mathrm{phys}}$ requires $H = \lceil Rh_{\mathrm{phys}} \rceil$ in the normalized first-event controller, so its bill includes the aggregate upper update rate $R = 2Nr^*$. The finite quantized time conversion uses that same calibrated time unit.

For scale, a single operating update at $\theta = 2^{-20}$ and $\rho = 2$ has the sufficient values $m = 23$, $n = 1656$, $\beta = \theta/(8n)$, $D = 19.5$ in the model time unit (or 20 after upward whole-unit rounding), and $\Delta \le \beta/(256\rho)$, approximately $1.41 \times 10^{-13}$. It may require 1656 focal pins and 32292 units of censored observation time, or 33120 with that upward rounding, before pin, settling, context and timer costs. The sufficient deadline-uniform grid has 109 bits using the rational bound $e < 3$. The precise finite constants are conservative sufficient costs. They do not demonstrate an economical device or a universal lower bound. At small true $\lambda$ this bill remains independent of $1/\lambda$, but the required finite precision and pin services remain real.

## 7 Safe preparation and a consequential finite-menu decision

For a physical mission starting at $\nu_u = \pi^u(\cdot \mid A)$, use the accepted repairable-chain preparation and capped safe inspections with its proved error budget and actual operating exposure. The mission clock starts only after the accepted safe inspection completes. A simulator initial draw can instead use experimental Gibbs sampling followed by safe rejection; with a certified safe mass $a_0$ its cost pays $a_0^{-1}$, and a finite rejection cap and a safe fallback give the declared TV error. For the inherited equilibrium-safe physical preparation, require $\pi^u(A) \ge 1/2$ for *every* menu item to be prepared; this condition is certified or supplied separately for each item. If a different preparation is claimed with smaller safe mass, its exposure and rejection cost must explicitly pay the appropriate inverse safe mass. These are separate physical and computational initializations.

The intervention is chosen before observing the realized accepted initial configuration. Preparation after the chosen intervention, its field ramp, settling and faithful reproduction of the chosen operating environment are priced. The state of the live system during potentially long acquisition is not protected by this theorem; paid safe shutdown, a continuing preparation protocol, or its actual exposure risk must be specified.

**Corollary 7.1** (Acquired bounded-loss intervention choice)**.** *Let $K = |\mathcal{U}| < \infty$, and let $\ell_u$ be a computable loss of the returned record in $[0, 1]$. It may combine time, terminal damage subtype, precursor attributes and a known scaled intervention cost. Suppose each compiled menu generator has uniform conditional record error at most $\varepsilon_s$, on a joint shared event of probability at least $1 - \delta_{\mathrm{cal}}$. For $a > 0$, use*

$$n_{\mathrm{dec}} \ge \frac{1}{2a^2} \log \frac{2K}{\delta_{\mathrm{dec}}}$$

*fresh generated records per menu element, and choose an element minimizing its empirical mean loss. If each computed loss has absolute error at most $\varepsilon_{\mathrm{eval}}$, then, with probability at least $1 - \delta_{\mathrm{cal}} - \delta_{\mathrm{dec}}$,*

$$L_{\widehat{u}} \le \min_{u \in \mathcal{U}} L_u + 2(a + \varepsilon_s + \varepsilon_{\mathrm{eval}}), \qquad L_u = \mathbb{E}_{\nu_u, Q^u} \ell_u.$$

*If actual deployment-preparation laws are within TV $\kappa_{\mathrm{prep}}$ of the corresponding $\nu_u$, the bound relative to the best similarly prepared menu mission adds $2\kappa_{\mathrm{prep}}$.*

*Proof.* Condition on a good shared calibration state. For each record, its conditional expected loss differs from $L_u$ by at most $\varepsilon_s$. Losses minus these conditional means form bounded martingale differences. Conditional Hoeffding gives a two-sided deviation at most $a$ with failure probability $2e^{-2n_{\mathrm{dec}}a^2}$. A union bound over the $K$ menu elements and the loss-evaluation error bound imply that every empirical mean differs from its true $L_u$ by at most $a+\varepsilon_s+\varepsilon_{\mathrm{eval}}$. Comparing the empirical minimizer to a true minimizer gives twice that error. A bounded loss changes by at most the TV error of its initial law; apply this once to the selected and once to the comparator mission. □

The complete decision bill is $K$ menu compilations, $Kn_{\mathrm{dec}}$ fresh record draws with (10), plus actual chosen-menu deployment and safe preparation. Context/environment sharing may lower that bill only when physically justified. The result is additive regret for a finite menu, not relative accuracy for a tiny conditional subtype, unbounded lifetime loss, arbitrary adaptive policy optimization, or an unobserved natural failure mechanism. For a tiny deadline risk, the separate acquired relative whole-CDF construction may be the appropriate decision interface.

# 8 What this earns, prior mechanisms, and preserved boundaries

The new operational consequence is a resource choice for an acquired joint first-event law. A fitted-model route can learn numerical energy and kinetic tables and use a marked-law stability theorem. The present route instead keeps calling faithful local cells to implement the true kernels; it need not learn a fine operating-rate table, and under the pairwise coarse certificate need not enumerate that table's $2^d$ contexts at all. The optional fitted-static hybrid learns only the invariant weights finely and supplies true kinetics experimentally. These routes support the same finite-menu bounded-loss endpoint with different acquisition, per-record service and metrology costs. No one route is universally cheaper, and none implies that a natural system's closed model has been identified from short observations.

Bernoulli factories already transform samples from unknown probabilities into samples of functions of those probabilities; analytic-factory constructions and their finite tail theory are established [2]. Two-coin methods for intractable transition probabilities and Bernoulli races are likewise established [3, 4]. The wider idea that sampling can replace exact numerical evaluation is also present in exact-approximate and pseudo-marginal MCMC [5] and the exchange algorithm [6]. Thus a generic slogan that parameters need not be estimated before sampling is not the contribution.

The specific construction here turns the *permitted local physical interventions* into the static and dynamic interfaces of a finite rare-first-event operation. It closes four concrete interfaces:

1. A balanced artificial-field observation becomes an operating Gibbs conditional, while paying the actual field shift and finite relaxation.

2. An actual operating first-exit race becomes a uniformized transition coin, without equating biased and operating clocks or paying a rare full dwell. Arbitrary locally requested state pins and timing are charged.

3. Unknown terminal weights in heavy-prefix counting are replaced by a path of measured local log odds, so the rare static interface is earned.

4. Conditional query errors and conditional means are propagated through static sampling, Palm calibration, marked rejection and decisions; physical iid records are not assumed.

The accepted cut and Palm operations supply the global rarity removal; this construction does not claim to replace their strong static and mixing premises with mere local observations. No historical firstness is asserted for the combined theorem.

### 8.1 Useful failed routes and why they are not used

1. Coupling two full lifetime trajectories step by step accumulates error over an enormous waiting time. It does not give the present no-inverse-rarity operation. The completed finite block sampler skips that lifetime, while the present procedure implements its short queries.

2. A bounded-time passive record cannot relatively estimate an arbitrarily small local exit rate. The censored-exit factory does not estimate that rate; it samples a coin whose correspondingly rare outcome has the correct probability. The externally priced state pin is essential.

3. An artificial balancing field may change the total relaxation clock. Using that observed clock as the operating clock would be wrong. Every operating coin here uses the actual requested operating environment.

4. Coarse energy learning does not make field precision free. Unbiasing with $e^{u-z}$ requires accurate knowledge and reproduction of the actual additive shift. A hardware error floor yields an output error floor unless another interface is proved.

5. Fast Gibbs sampling alone gives additive estimates of rare event occupancy. Relative pin masses require the heavy-prefix identity and the measured weight-ratio path. Counting a cut by waiting for it would restore inverse rarity.

6. Reusing past random bits or observed cell states as though they were independent fresh outputs is unjustified. Uniform conditional kernel bounds, holds and conditional-mean concentration are the repairs.

7. Coupling every high-level calibration call at error $\delta/D$ can force a polynomial inverse-confidence static bill. The conditional mean lemma retains a fixed per-proposal bias budget while sample counts carry $\log(1/\delta)$. The former wrapper is a valid but weaker option.

### 8.2 Execution status

The companion `factory_audit.py` executes exact rational finite-walk and two-coin-factory audits, including extremely small operating-rate examples. It checks finite success probabilities against the target identities and rigorous truncation bounds, and records the finite service parameters in `factory_results.json`. The executed audit has 10 exact finite-walk cases, 20 exact Gibbs-factory cases and 4 independently enumerated pin-mass identities. All checks passed. It includes a positive operating rate $2^{-100}$ and, separately, a zero-rate boundary diagnostic outside the positive-clock theorem. These checks exercise the two local factories and their finite caps. They do not execute the full interacting cut enumerator, the full physical first-event controller, hardware calibration, or an actual experiment. The theorem and its resource bill, rather than a small simulation, earn the general consequence.

Every accepted original remains unchanged. The earlier acquired relative CDF, independent first-event record sampler and finite-bit clock, active acquisition/preparation/prevention, fixed-law prediction and fixed-mechanics thermal operations remain at their original scopes. The earlier all-strength quantum laws and thermal effusion result are not reinterpreted by this classical construction.

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